

Deep Research: Avatrada 57 Topic 027

Engineering Report: FINRA Rule 2360 Delta Aggregation

TO: Engineering & Compliance Teams **FROM:** Autonomous Technical Researcher **DATE:** October 26, 2023 **SUBJECT:** Deep-Dive on FINRA Rule 2360 Delta-Based Position Limit Calculation

1.0 Technical Deconstruction

This section deconstructs the mechanisms and logic behind the delta-based aggregation method for FINRA Rule 2360 position limits.

1.1 Objective of FINRA Rule 2360

FINRA Rule 2360 establishes position limits on equity options to prevent any single entity or group from establishing an excessively large position that could be used to manipulate the market for the underlying security. Compliance is traditionally measured by counting the total number of contracts on the same "side of the market" (e.g., long calls and short puts are bullish; long puts and short calls are bearish).

However, FINRA provides a more sophisticated alternative: the **Delta-Based Method**. This method acknowledges that not all options carry the same market risk. An at-the-money option has a much larger impact on the underlying than a deep out-of-the-money option. The delta-based approach converts all option positions into a "Delta-Equivalent Share Count," which represents the portfolio's net equivalent long or short share position in the underlying security.

1.2 The Core Aggregation Formula

The "Delta-Equivalent Share Count" is the net sum of the delta-adjusted values of all option positions on a single underlying security. The formula for this calculation is:

$$\text{Delta-Equivalent Share Count} = \sum (\Delta_i * C_i * M)$$

Where: * Σ (Sigma) represents the summation across all individual option positions (legs) in the portfolio for a given underlying. * Δ_i (Delta) is the delta of the specific option leg i . * Long Calls have a delta between 0 and +1. * Short Calls have a delta between 0 and -1. * Long Puts have a delta between -1 and 0. * Short Puts have a delta between 0 and +1. * C_i (Contracts) is the number of contracts for leg i . This value is **positive for long positions** and **negative for short positions**. * M (Multiplier) is the number of shares per contract, which is typically **100** for standard US equity options.

1.3 Example Calculation: Mixed SPY Portfolio

Let's deconstruct a hypothetical portfolio of SPY options to calculate its single Delta-Equivalent Share Count.

Portfolio Positions: 1. **Long 2,000 SPY 450 Calls** (Bullish Spread Leg) 2. **Short 2,000 SPY 460 Calls** (Bearish Spread Leg) 3. **Long 500 SPY 440 Puts** (Bearish Position) 4. **Short 1,000 SPY 430 Puts** (Bullish Position)

Assumed Market Data & Deltas: * SPY Underlying Price: \$452.00 * Position 1 (Long 2000 C @ 450): Delta (Δ) = **+0.65** * Position 2 (Short 2000 C @ 460): Delta (Δ) = **+0.35** (Note: The option's delta is positive, but the position is short). * Position 3 (Long 500 P @ 440): Delta (Δ) = **-0.20** * Position 4 (Short 1000 P @ 430): Delta (Δ) = **-0.10** (Note: The option's delta is negative, but the position is short).

Calculation Steps:

1. Leg 1 (Long Calls):

- $\Delta_1 = +0.65$

- $C_1 = +2000$ (Long position)
- Share Count = $0.65 * 2000 * 100 = +130,000$

2. Leg 2 (Short Calls):

- $\Delta_2 = +0.35$
- $C_2 = -2000$ (Short position)
- Share Count = $0.35 * (-2000) * 100 = -70,000$

3. Leg 3 (Long Puts):

- $\Delta_3 = -0.20$
- $C_3 = +500$ (Long position)
- Share Count = $-0.20 * 500 * 100 = -10,000$

4. Leg 4 (Short Puts):

- $\Delta_4 = -0.10$
- $C_4 = -1000$ (Short position)
- Share Count = $-0.10 * (-1000) * 100 = +10,000$

Final Aggregation:

Total Delta-Equivalent Shares = $130,000 - 70,000 - 10,000 + 10,000$
 Total Delta-Equivalent Shares = $+60,000$

This portfolio has a net Delta-Equivalent Share Count of **60,000 long shares**. This final number is what would be compared against the FINRA position limit for SPY, which is expressed in shares for firms using the delta-based method.

1.4 Application to a Principal's Personal Account

FINRA rules are designed to prevent circumvention by spreading positions across multiple accounts. The key concept is **control and acting in concert**.

- **Aggregation is Required:** A Principal acting as a sole proprietorship is, for regulatory purposes, indistinct from their firm. Any accounts over which

the Principal has direct or indirect control, including personal brokerage accounts, must be aggregated with the firm's accounts.

- **Rationale:** The rule applies to a "person," which includes any entity, sole proprietorship, or individual. If the Principal's personal account and the firm's proprietary account both hold SPY options, they are considered to be under common control.
- **Conclusion:** The positions in the Principal's personal account **must be included** in the firm's total Delta-Equivalent Share Count calculation for SPY. Failure to do so would be a serious violation of Rule 2360.

2.0 Implementation Strategy

Building a robust, real-time system for this calculation requires integrating position data, market data, and a financial modeling library.

2.1 System Architecture & Data Flow

1. **Position Ingestion:** The system must have access to a real-time feed of all option positions from all relevant accounts (firm prop, principal's personal, etc.). This is typically sourced from a portfolio management system or a direct drop-copy from the executing broker.
2. **Market Data Service:** A low-latency market data provider is needed for:
 - Underlying asset price (e.g., SPY).
 - Implied Volatility (IV) for each option series.
 - Risk-free interest rate.
 - Dividend yield for the underlying.
3. **Delta Calculation Engine:** This core component takes the position and market data to calculate the delta for each leg.
4. **Aggregation & Limit Check:** The engine sums the delta-equivalent shares and compares the absolute value against the pre-configured FINRA limit for the specific symbol.
5. **Alerting & Reporting:** If a pre-defined threshold (e.g., 90% of the limit) or the limit itself is breached, the system must generate immediate alerts to the trading and compliance desks.

2.2 Technology Stack (Python Example)

Python is well-suited for this task due to its powerful financial and data analysis libraries.

- **Libraries:**

- `pandas` : For managing and manipulating position data in a structured DataFrame.
- `py_vollib` : A standard, open-source library for calculating option greeks, including delta, based on the Black-Scholes-Merton model.
- `requests` or a dedicated SDK: For fetching data from market data provider APIs (e.g., Polygon.io, Bloomberg).

- **Execution Environment:** Can be run as a scheduled job (e.g., every minute) or a streaming application for real-time monitoring.

2.3 Pseudocode & Python Snippet

Here is a Python function demonstrating the core logic for the SPY portfolio example.

```
import pandas as pd
from py_vollib.black_scholes import black_scholes

# Note: In a real system, 'greeks' would be calculated using a library
# that takes live market data (price, vol, rate) as input.
# For this example, we use the pre-determined deltas.

def calculate_delta_equivalent_shares(portfolio_df: pd.DataFrame) -> float:
    """
    Calculates the total Delta-Equivalent Share Count for a portfolio.

    Args:
        portfolio_df: A pandas DataFrame with columns:
            ['symbol', 'type', 'strike', 'contracts', 'delta']
            'contracts' is positive for long, negative for short.

    Returns:
```

```

        The total delta-equivalent share count.
    """
    OPTION_MULTIPLIER = 100

    # Vectorized calculation for efficiency
    portfolio_df['delta_shares'] = (
        portfolio_df['delta'] *
        portfolio_df['contracts'] *
        OPTION_MULTIPLIER
    )

    total_delta_equivalent_shares = portfolio_df['delta_shares'].sum()

    return total_delta_equivalent_shares

# 1. Define the portfolio from our example
data = {
    'symbol': ['SPY', 'SPY', 'SPY', 'SPY'],
    'type': ['Call', 'Call', 'Put', 'Put'],
    'strike': [450, 460, 440, 430],
    'contracts': [2000, -2000, 500, -1000], # Note the negative sign for shorts
    'delta': [0.65, 0.35, -0.20, -0.10]
}
portfolio = pd.DataFrame(data)

# 2. Run the calculation
net_delta_shares = calculate_delta_equivalent_shares(portfolio)

# 3. Output the result
print(f"Portfolio DataFrame:\n{portfolio}\n")
print(f"Total Delta-Equivalent Share Count: {net_delta_shares:,.0f}")

# Example Limit Check
SPY_POSITION_LIMIT = 900000 # Example limit
if abs(net_delta_shares) > SPY_POSITION_LIMIT:
    print("\n*** ALERT: POSITION LIMIT BREACHED ***")
else:
    print(f"\nPosition is within the {SPY_POSITION_LIMIT:,} share limit.")

```

3.0 Critical Analysis

While powerful, the delta-based method introduces complexities and potential points of failure that must be managed.

3.1 Potential Failure Modes

- **Stale Market Data:** The entire calculation is highly sensitive to the inputs, especially the underlying price and implied volatility. A stale data feed will produce an inaccurate delta, leading to a "phantom" compliance status. The system must have data health checks and fail-safes if a data feed becomes latent.
- **Model Risk (Black-Scholes):** The Black-Scholes model has known limitations. It assumes constant volatility and risk-free rates, which is not true in reality. For American-style options (most equity options), a model like Binomial/Trinomial pricing is technically more accurate, especially around dividends. However, Black-Scholes is the industry standard for its speed and simplicity in this context.
- **Data Aggregation Failure:** The system must be architected to guarantee that positions from *all* required accounts are included. A failure to connect to a data source for one account (e.g., the Principal's personal account) should trigger a system-wide alert, as the resulting calculation would be incomplete and non-compliant.

3.2 Edge Cases

- **Gamma Risk at Expiration:** As options near expiration, their delta can change extremely rapidly with small moves in the underlying price (high gamma). A portfolio that is compliant mid-day could rapidly breach its limit near the 4:00 PM ET close. Monitoring frequency must be increased significantly for near-expiry positions.
- **Illiquid Options:** For deep in-the-money or far out-of-the-money options, the calculated delta from a model might not reflect the option's true hedging reality. A deep ITM call's delta may approach 1.0, effectively making it a stock-equivalent position.

- **Corporate Actions:** Stock splits, mergers, or special dividends require careful handling. A standard 2-for-1 split would double the number of shares and halve the strike, but the option multiplier (100) might change to reflect a non-standard deliverable. The system must correctly ingest and apply these adjustments.

3.3 Optimizations

- **Vectorization:** As shown in the Python example, performing calculations on the entire portfolio array at once (vectorization) using libraries like NumPy and Pandas is orders of magnitude faster than iterating through positions one by one.
- **Caching:** Market data inputs (especially IV and rates) do not change tick-by-tick. Caching these values for a short duration (e.g., 1-5 seconds) can significantly reduce the load on market data services without sacrificing meaningful accuracy.
- **Tiered Alerting:** Implement multiple alert thresholds. For example:
 - **Level 1 (80% of limit):** Informational alert to the trading desk.
 - **Level 2 (95% of limit):** High-priority warning to trading and compliance.
 - **Level 3 (100% breach):** Critical, "cease trading" alert and automated reporting. This allows for proactive risk management rather than reactive damage control.